① Fie $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid x-y-z=0 \right\}$

$U = Sp\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}; \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \right\}$ Determinați $V, U, U+V, U\cap V$ (bază + dim)

a) $x-y-z=0$

Fie $y=\alpha$, $\alpha\in\mathbb{R}$; $z=\beta$, $\beta\in\mathbb{R}$ $\Rightarrow x=\alpha+\beta \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \alpha+\beta \\ \alpha \\ \beta \end{pmatrix} = \alpha\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \beta\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

$V = \left\{ \underbrace{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}_{v_1}\alpha + \underbrace{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}}_{v_2}\beta ; \alpha,\beta\in\mathbb{R} \right\}$

$V = Sp\{v_1; v_2\} \Rightarrow V =$ bază

(v_1, v_2 sunt obținuți dintr-un sistem $\Rightarrow$ sunt liniari independenți)

$\dim_{\mathbb{R}} V = 2$

b) $U = Sp\left\{ \underbrace{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}_{u_1}; \underbrace{\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}}_{u_2} \right\}$

u_1, u_2 liniari independenți (pt. că nu sunt proporționali)

$\Rightarrow U =$ bază

$\dim_{\mathbb{R}} U = 2$

c) $U+V = Sp\{u_1, u_2, v_1, v_2\} = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}; \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}; \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}; \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$

$\dim_{\mathbb{R}} U+V \le \dim \mathbb{R}^3 = 3$

$\dim_{\mathbb{R}} U+V \ge \max(\dim_{\mathbb{R}} V, \dim_{\mathbb{R}} U) = 2$ $\Rightarrow$

$\Rightarrow 2 \le \dim_{\mathbb{R}} U+V \le 3$

Verif lin. indep. u_1, u_2, v_2

$\begin{vmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 2+1 = 3 \ne 0 \Rightarrow$ vect. lin. indep. $\Rightarrow$

$\Rightarrow U+V$ bază

$U+V = Sp\{u_1, u_2, v_2\}$

$\dim_{\mathbb{R}}(U+V) = 3$

d) F. Grassmann

$\underset{3}{\dim_{\mathbb{R}}(U+V)} = \underset{2}{\dim_{\mathbb{R}} U} + \underset{2}{\dim_{\mathbb{R}} V} - \dim_{\mathbb{R}}(U\cap V)$

$\dim_{\mathbb{R}}(U\cap V) = 1$

$U\cap V = \{w\in\mathbb{R}^3 \mid w\in U$ și $w\in V\}$

$w\in U \Rightarrow (\exists)\ \alpha_1,\alpha_2\in\mathbb{R}$ a.î. $w = \alpha_1 u_1 + \alpha_2 u_2$

$w\in V \Rightarrow (\exists)\ \beta_1,\beta_2\in\mathbb{R}$ a.î. $w = \beta_1 v_1 + \beta_2 v_2$

$\alpha_1 u_1 + \alpha_2 u_2 = \beta_1 v_1 + \beta_2 v_2$

$\alpha_1\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \alpha_2\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} = \beta_1\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \beta_2\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \Rightarrow \begin{cases} \alpha_1 = \beta_1+\beta_2 \\ \alpha_1+2\alpha_2 = \beta_1 \\ \alpha_2 = \beta_2 \end{cases} \Rightarrow$

$\Rightarrow \begin{cases} \alpha_1 = \beta_1+\beta_2 \\ \alpha_1 = \beta_1-2\beta_2 \\ \alpha_2 = \beta_2 \end{cases} \Rightarrow 0 = 3\beta_2 \Rightarrow \beta_2 = 0 \Rightarrow \alpha_2 = 0 \Rightarrow \alpha_1 = \beta_1 = \gamma$, $\gamma\in\mathbb{R}$

$\begin{cases} \alpha_1 = \gamma,\ \gamma\in\mathbb{R} \\ \alpha_2 = 0 \end{cases} \Rightarrow w = \gamma u_1 + 0u_2 = \gamma u_1$

$\begin{cases} \beta_1 = \gamma,\ \gamma\in\mathbb{R} \\ \beta_2 = 0 \end{cases} \Rightarrow w = \gamma v_1 + 0v_2 = \gamma v_1$

$\Rightarrow w = \gamma\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \Rightarrow$

$\Rightarrow U\cap V = Sp\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}$

② $V = Sp\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}; \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}; \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix} \right\}$ $U = Sp\left\{ \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}; \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}; \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix} \right\}$

Determinați $V, U, V+U, U\cap V$ (bază + dim)

a) $V = Sp\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}; \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}; \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix} \right\}$

$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 3 \end{vmatrix} = \cancel{3} + \cancel{3} - 2 - 1 + 3 - \cancel{6} = 0$

$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}; \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ lin. ind. pt. că nu sunt proporționali $\Rightarrow$

$\Rightarrow v_1, v_2$ lin ind $\Rightarrow V = Sp\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}; \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\} \Rightarrow V =$ bază

$\dim_{\mathbb{R}} V = 2$

b) $U = Sp\left\{ \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}; \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}; \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix} \right\}$

$\begin{vmatrix} 2 & 1 & 1 \\ 3 & 2 & 1 \\ -1 & 2 & -3 \end{vmatrix} = -12 - 1 + 6 + 2 - 4 + 9 = -17 + 17 = 0$

$\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}; \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ nu sunt prop. $\Rightarrow u_1, u_2$ lin. indep. $\Rightarrow$

$\Rightarrow U = Sp\left\{ \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}; \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right\}$ bază

$\dim_{\mathbb{R}} U = 2$

c) $U+V = \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}; \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}; \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}; \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right\}$

$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 2 & 1 & 3 & 2 \\ 1 & -1 & -1 & 2 \end{pmatrix} \xrightarrow[L_3-L_1]{L_2-2L_1} \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & -2 & -3 & 1 \end{pmatrix} \xrightarrow{L_3-2L_2} \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}$

$U+V = Sp\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}; \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}; \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} \right\}$ bază

$\dim_{\mathbb{R}^3}(U+V) = 3$

d) $U\cap V = \{w\in\mathbb{R}^3 \mid w\in U$ și $w\in V\}$

$w = \underbrace{\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}}_{v_1}\alpha_1 + \underbrace{\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}}_{v_2}\alpha_2 \Rightarrow \begin{cases} \alpha_1+\alpha_2 = 2\beta_1+\beta_2 \\ 2\alpha_1+\alpha_2 = 3\beta_1+2\beta_2 \\ \alpha_1-\alpha_2 = -\beta_1+2\beta_2 \end{cases}$

$w = \overbrace{\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}}^{=u_1}\beta_1 + \overbrace{\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}}^{=u_2}\beta_2$

$\begin{pmatrix} 1 & 1 & -2 & -1 \\ 2 & 1 & -3 & -2 \\ 1 & -1 & 1 & -2 \end{pmatrix} \xrightarrow[L_3-L_1]{L_2-2L_1} \begin{pmatrix} 1 & 1 & -2 & -1 \\ 0 & -1 & 1 & 0 \\ 0 & -2 & 3 & -1 \end{pmatrix} \xrightarrow{L_3-2L_2} \begin{pmatrix} 1 & 1 & -2 & -1 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$

$\begin{cases} \alpha_1+\alpha_2-2\beta_1-\beta_2 = 0 \\ -\alpha_2+\beta_1 = 0 \\ \beta_1-\beta_2 = 0 \end{cases}$ sist. comp. nedet. cu o necunoscută secundară $\Rightarrow \dim_{\mathbb{R}^3}(U\cap V) = 1$

$\beta_1 = \beta_2 = \alpha_2 = \gamma$; $\gamma\in\mathbb{R}$

$\alpha_1 + \cancel{\gamma} - 2\gamma - \cancel{\gamma} = 0 \Rightarrow \alpha_1 = 2\gamma$

$\begin{cases} \alpha_1 = 2\gamma \\ \alpha_2 = \gamma \end{cases}$ $\gamma\in\mathbb{R} \Rightarrow w = 2\gamma v_1 + \gamma v_2 = 2\gamma\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \gamma\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = \gamma\begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix}$

$\begin{cases} \beta_1 = \gamma \\ \beta_2 = \gamma \end{cases}$ $\gamma\in\mathbb{R} \Rightarrow w = \gamma u_1 + \gamma u_2 = \gamma\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} + \gamma\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = \gamma\begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix}$

$U\cap V = Sp\left\{ \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} \right\}$ $\dim_{\mathbb{R}^3}(U\cap V) = 1$

③ Verificați $v\in W$

a) $v = \begin{pmatrix} -3 \\ 8 \\ 1 \end{pmatrix}$ $W = \left\{ \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}; \begin{pmatrix} 5 \\ -13 \\ -3 \end{pmatrix} \right\}$

Fie $\alpha,\beta\in\mathbb{R}$ a.î. $v = \alpha\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + \beta\begin{pmatrix} 5 \\ -13 \\ -3 \end{pmatrix}$

$\begin{cases} \alpha+5\beta = -3 \\ -2\alpha-13\beta = 8 \\ 3\alpha-3\beta = 1 \end{cases} \Rightarrow -16\beta-5\beta = 1+8+3 \Rightarrow$

$-21\beta = 12 \Rightarrow \beta = -\frac{12}{21} = -\frac{4}{7}$

$\alpha = -3-5\beta = -3+\frac{20}{7} = \frac{-21+20}{7} = -\frac{1}{7}$

$\alpha = \frac{1+3\beta}{3} = \frac{1}{3} - \frac{4}{7} = \frac{7-12}{21} = \frac{-5}{21}$ $\not\Rightarrow$ $\nexists$ sist. incomp.

b) $v = 1-4x+6x^2$

$W = Sp\{1-x+x^2, 2+x-3x^2\}$

$v\in W \Rightarrow (\exists)\ \alpha,\beta\in\mathbb{R}$ a.î. $v = \alpha(1-x+x^2)+\beta(2+x-3x^2)$

$\Rightarrow v = (\alpha+2\beta)+(-\alpha+\beta)x+(\alpha-3\beta) \Rightarrow \begin{cases} \alpha+2\beta = 1 \\ -\alpha+\beta = -4 \\ \alpha-3\beta = 6 \end{cases}$

$\Rightarrow -2\beta = 2 \Rightarrow \beta = -1 \Rightarrow \alpha = 6+3\beta = 6-3 = 3$

$3-2=1$ adev. $\Rightarrow v\in W$

c) $v = \sin 2t$ $W = Sp\{\sin t, \cos t\}$

$v\in W \Rightarrow (\exists)\ \alpha,\beta\in\mathbb{R}$ a.î. $v = \alpha\sin t + \beta\cos t = \sin 2t$

Pt $t=0 \Rightarrow \beta = 0$

Pt $t=\frac{\pi}{2} \Rightarrow \alpha = \sin\pi = 0$

$\sin 2t = 0\sin t + 0\cdot\cos t$ $(\forall)\ t\in\mathbb{R}$ Fals $\Rightarrow v\notin W$

d) $v = \begin{pmatrix} -4 \\ 0 \\ -3 \end{pmatrix}$ $W = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in\mathbb{R}^3 \;\middle|\; \begin{cases} 3x+5y-4z = 0 \\ -3x-2y+4z = 0 \\ 6x+y-8z = 0 \end{cases} \right\}$

Verificăm dacă v este soluție

$3(-4)+5\cdot0-4(-3) = -12+12 = 0$

$-3\cdot(-4)-2\cdot0+4(-3) = 12-12 = 0$

$6\cdot(-3)+0-8(-3) = (-3)\cdot(6-8) \ne 0$

$\not\Rightarrow v\notin W$

④ Verificați dacă B este o bază pt. subsp. pe care-l generează și calculați coord. vect. v. în această bază

a) $B = \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}; \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$ $v = \begin{pmatrix} 1 \\ 6 \\ 2 \end{pmatrix}$

v_1, v_2 nu sunt proporționali $\Rightarrow v_1, v_2$ lin. ind. $\Rightarrow B$ bază

$[v]_B = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \Rightarrow \begin{pmatrix} 1 \\ 6 \\ 2 \end{pmatrix} = \alpha\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \beta\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \Rightarrow \begin{cases} \alpha+\beta = 1 \quad (A) \\ 2\alpha = 6 \Rightarrow \alpha = 3 \\ -\beta = 2 \Rightarrow \beta = -2 \end{cases}$

$\Rightarrow [v]_B = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$

b) $B = \{1+x, x+x^2, 1+x^2\}$

$v = 1+2x-x^2$

$\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 1+1 = 2 \ne 0 \Rightarrow v_1, v_2, v_3$ lin. indep $\Rightarrow$

$\Rightarrow B$ bază

$[v]_B = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} \Rightarrow 1+2x-x^2 = \alpha(1+x)+\beta(x+x^2)+\gamma(1+x^2)$

$\Rightarrow \begin{cases} \alpha+\gamma = 1 \\ \alpha+\beta = 2 \\ \beta+\gamma = -1 \end{cases} \Rightarrow \beta-\gamma = 1$ $\Rightarrow 2\beta = 0 \Rightarrow \beta = 0$, $\gamma = -1$, $\alpha = 2$ $\Rightarrow [v]_B = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$

⑤ $B_1 = \left\{ \underset{u_1}{\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}}; \underset{u_2}{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}}; \underset{u_3}{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}} \right\}$ — baze în $\mathbb{R}^3$

$B_2 = \left\{ \underset{v_1}{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}; \underset{v_2}{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}; \underset{v_3}{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}} \right\}$

$v\in\mathbb{R}^3$, $[v]_{B_1} = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$

$P_{B_1}^{B_2}$; $[v]_{B_2} = ?$

$P_{B_1}^{B_2}$ $\quad P_{B_1}^{B_2} = \left([v_1]_{B_1} \mid [v_2]_{B_1} \mid [v_3]_{B_1} \right)$

$(u_1\ u_2\ u_3 \,\vdots\, v_1\ v_2\ v_3) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 1 & 1 \\ 2 & 1 & 0 & 1 & 1 & 0 \\ 4 & 1 & 1 & 1 & 0 & 1 \end{array}\right) \longrightarrow$

$\xrightarrow[L_3-4L_1]{L_2-2L_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & -1 & -1 & -2 \\ 0 & 1 & -3 & -3 & -4 & -3 \end{array}\right) \xrightarrow{L_3-L_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & -1 & -1 & -2 \\ 0 & 0 & -1 & -2 & -3 & -1 \end{array}\right)$

$\xrightarrow{(-1)\cdot L_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & -1 & -1 & -2 \\ 0 & 0 & 1 & 2 & 3 & 1 \end{array}\right) \xrightarrow[L_1-L_3]{L_2+2L_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -2 & 0 \\ 0 & 1 & 0 & 3 & 5 & 0 \\ 0 & 0 & 1 & 2 & 3 & 1 \end{array}\right)$

$P_{B_1}^{B_2} = \begin{pmatrix} -1 & -2 & 0 \\ 3 & 5 & 0 \\ 2 & 3 & 1 \end{pmatrix}$

$[v]_{B_2}$ $\quad [v]_{B_2} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$; $[v]_{B_1} = P_{B_1}^{B_2}[v]_{B_2}$ $\Rightarrow \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 & -2 & 0 \\ 3 & 5 & 0 \\ 2 & 3 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$\begin{cases} -x-2y = -1 \\ 3x+5y = 2 \\ 2x+3y-z = 3 \end{cases} \xrightarrow{L_2+L_1-L_3} z = 2-1-3 = -2$

$\begin{cases} -3x-6y = -3 \\ 3x+5y = 2 \end{cases} \Rightarrow -y = -1 \Rightarrow y = 1$

$-x-2 = -1 \Rightarrow -x = 1 \Rightarrow x = -1$

$[v]_{B_2} = \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix}$

$$\boxed{\begin{gathered} P_{B_1}^{B_2} = (B_1 \,\vdots\, B_2) \longrightarrow (I \,\vdots\, P_{B_1}^{B_2}) \\ [v]_{B_1} = P_{B_1}^{B_2}[v]_{B_2} \end{gathered}}$$